

Data Engineering Pre-Scoping Specification

High-Velocity Economic Layer (E_{macro} Input Layer) for Spatial Embedding Tensor

Executive Summary: All six external macro data sources successfully support county-level extraction across the 50 US States, satisfying the baseline geographic requirement ($N = 3,142$ counties and equivalents). Spatial compatibility is confirmed; however, significant variation in update frequencies (continuous to decennial) and reporting latencies requires an offline asynchronous batch ingestion strategy to isolate downstream applications from API bottlenecks. This document outlines boundaries and identifies core technical risks for pipeline plausibility validation.

Document Scope & Operational Assumptions:

- **Downstream Objective:** The target recipient is a unified spatial embedding tensor preserving localized economic velocity and global structural relationships.
- **Algorithmic Execution:** This pre-scoping stage documents boundaries and flags technical implementation risks without enacting final programmatic code solutions.

1. Core Source Ingestion & Validation Matrix

The following matrix evaluates the availability, temporal cadence, access mechanics, and primary systemic risks associated with each input macro source.

Source	County Data?	Update Frequency	Reporting Lag	Access Protocol	Primary Systemic Risk
A: Wikipedia Introductory Corpora	Yes (Text pattern matching)	Continuous (Real-time)	None	Wikimedia Enterprise HTML / Structured API	Token overflow (> 8,192 tokens); naming ambiguity
B: BLS Quarterly Census (QCEW)	Yes (5-digit FIPS files)	Quarterly	5 – 6 Months	HTTP Flat-File Server (data.bls.gov/cew/)	Proprietary data suppression (NaN cell blocks)
C: FRED Time-Series	Yes (LAUS / GDP series)	Monthly (LAUS) Annually (GDP)	~1 Month (LAUS) ~1 Year (GDP)	REST API via developer token (api.stlouisfed.org)	Strict rate limit thresholds (120 req/min); IP bans
D: BTS FAF5 Flows	Yes (Experimental County Matrix)	Intermittent / Multi-Year	1 – 2 Years	Bulk HTTPS Data Download (bts.gov)	Combinatorial matrix explosion ($N \times N$ dense network edges)
E: IRS Statistics of Income	Yes (Aggregated Form 1040)	Annually	18 – 24 Months	Bulk CSV Zip Archives (irs.gov/statistics)	Upstream schema mutation (tax form line item shifts)
F: USDA Economic Research	Yes (Universal mapping)	Decennial (Every 10 years)	N/A (Static Anchor)	Static Tabular Ingestion (ers.usda.gov)	None. Acts as a deterministic baseline anchor

2. Individual Source Scoping & Boundary Actions

Source A: Wikipedia Introductory Corpora (Narrative Identity)

Operational Boundary: Ingests the introductory text sections using the structured pattern string "[*County Name*] County, [*State Name*]".

Edge Case Handling: Independent cities lacking standard county parent structures (e.g., St. Louis, MO; Baltimore, MD; Carson City, NV) pass through a regex translation table to map directly to independent FIPS codes (e.g., FIPS 29510 vs. FIPS 29189).

Token Window Limit Strategy: The downstream bge-m3 embedding model enforces an upper limit of 8,192 tokens. If an article length exceeds this threshold, a recursive character text splitter with 10% overlap is triggered, followed by a mean-pooling step across the generated chunks:

$$v_{\text{county}} = (1 / M) \times \sum_{m=1}^M \text{Encoder}(\text{text}_{\text{chunk}, m})$$

Identified Risk / Pushback: Naming patterns fail for Parishes (LA) and Boroughs (AK). Additionally, uniform mathematical mean-pooling dilutes high-signal economic narratives by weighing late-stage text blocks (e.g., lists of notable residents) equally with introductory paragraphs.

Source B: BLS Quarterly Census of Employment and Wages (Industrial Core)

Operational Boundary: Targets pre-calculated Location Quotients ($LQ_{c,i}$) across the 20 primary aggregate 2-digit NAICS codes for all 3,142 counties.

Data Suppression Resolution: To comply with federal privacy guidelines, the BLS suppresses data cells in counties where small employer counts could expose individual company operations.

Pre-Ingestion Action: The pipeline executes an Iterative Proportional Fitting (IPF) matrix completion loop. When an LQ cell is omitted, the missing value is estimated by solving an optimization problem matching known state-level distributions and total county constraints:

$$\min \sum_i \hat{E}_{c,i} \ln(\hat{E}_{c,i} / P_{s,i}) \quad \text{s.t.} \quad \sum_i \hat{E}_{c,i} = \text{Total County Employment}$$

Identified Risk / Pushback: Executing this formulation independently per county ignores multi-county column constraints. Missing values will fail to sum up to known state-level industry totals, introducing severe conservation-of-mass errors across the spatial tensor.

Source C: FRED Time-Series Slope Derivatives (Economic Velocity)

Operational Boundary: Pulls Local Area Unemployment Statistics (LAUS) and County GDP series to capture localized trend acceleration vectors.

API Rate-Limiting Strategy: Extracting two unique time-series across 3,142 counties requires 6,284 discrete requests. This requires an asynchronous worker queue wrapped in a token-bucket rate limiter restricted to 100 requests per minute.

Pre-Ingestion Action: To calculate the 3-year rolling first derivative ($\Delta y / \Delta t$) over the deseasonalized component, a continuous historical lookback window of 48 months is mandated.

Identified Risk / Pushback: A 48-month window provides exactly 12 months of padding for trend deseasonalization filters. This introduces severe end-of-period boundary distortions and tail effects on the final velocity derivative.

Source D: BTS FAF5 Experimental County Commodity Flows (Trade Gravity)

Operational Boundary: Evaluates inter-node flow of goods (millions of dollars, thousands of tons) across Standard Classification of Transported Goods (SCTG5) commodity definitions.

Combinatorial Density Control: Treating this network as a dense flat-file introduces an unmanageable input tensor of size $3,142 \times 3,142 \approx 9.87 \times 10^6$ potential matrix edges.

Pre-Ingestion Action: Compress incoming network density by implementing a top- K filter step. Retain only the top 5 destination FIPS codes by volume per origin, collapsing the long-tail into a single synthetic column labeled *External_Basin_Flow*.

Identified Risk / Pushback: Truncating the network to an asymmetric top-5 matrix destroys the global network topology. It severs macro-regional economic corridor identities, rendering the final embedding blind to network-layer trade gravity.

Source E: IRS Statistics of Income Panel Data (Capital Composition)

Operational Boundary: Extracts total adjusted gross income, W-2 wages, qualified dividends, and net capital gains (Form 1040 Schedule D) to compute the localized wealth asset-to-wage ratio (Φ_c).

Schema Mutation Strategy: Federal tax changes frequently rename, delete, or reorder line items across fiscal years.

Pre-Ingestion Action: Insert a structural abstraction layer between the raw CSV parser and the mathematical engine, mapping inputs to invariant conceptual strings (*total_w2_wages*, *net_cap_gains*) rather than tracking literal tax form line numbers.

Identified Risk / Pushback: Upstream mutations represent a binary structural failure point; while abstraction handles anticipated variation, any unmapped line shifts will cause pipeline halts requiring manual intervention.

Source F: USDA Economic Research Service (Structural Resilience Baseline)

Operational Boundary: Maps 2023 County Typology Codes to establish stable categorical baselines for rural and low-density micropolitan markets.

Vectorization Action: Convert static qualitative integers (e.g., *1 = Farming-dependent*) via a deterministic encoder into a fixed one-hot encoded array $\mathbf{v}_{USDA} \in \{0, 1\}^6$. This is intended to act as an unshifting spatial anchor.

Identified Risk / Pushback: One-hot encoding an input does not isolate its projection. If gradients flow freely through a joint neural layer, highly volatile time-series features from other sources will rotate the global coordinate spaces of these static anchors anyway.